Mohammad Uzair Khan

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Education

BITS Pilani, Dubai Campus — Dubai, UAE
B.E. Computer Science — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

Profile

Computer Science undergraduate in **AI solutions development**: **AI agents** with typed tool calling, retrieval and natural language inference, that turn pharmaceutical data into **actionable insights** for commercial teams. Self-driven and structured: every system is measured on held-out data against a baseline, with clear reporting of what worked and what did not.

AI Solutions & Data Science Projects

Brand Forecast Studio: sales forecasting for 19 MSD brands | *Python, LightGBM, Pydantic, TypeScript* — Live | Code
- Trained one global **LightGBM** model on **13,336** brand-quarters (535 brands) to forecast **19 MSD brands in 6 business units**; rolling-origin backtest: WAPE **17.9%** vs 19.4% for the best baseline, best method on 7 of 19 brands
- **Conformal calibration** raised P10–P90 coverage from 70% to **79.6%** on unseen quarters; one-command monitored pipeline with **14 health checks** (drift PSI, reconciliation) and a run log; Pydantic what-if and a **typed-tool assistant** (24/24 held-out questions)

Rx Market Pulse: AI analyst agent and forecasting | *Python, SQL, TypeScript, LLM tool calling* — Live | Code
- Built an **AI agent** answering commercial questions over **36.1M** Medicaid rows through **five typed tools** (spend, growth drivers, territories, forecasts, trials), with validated inputs, provenance and refusals; passed 20/20 curated real-model checks
- Backtested quarterly spend forecasts over **28 quarters** (chosen on 2022–23, tested on 2024–25); reported that the chosen method's 9.01% WAPE lost to the seasonal baseline's **5.54%**

SignalDesk AI: oncology competitive-intelligence agent | *Python, SQL, Gemini, TypeScript* — Live | Code
- Bounded **Gemini agent** with five typed evidence tools over **29** FDA announcements, 8 formulations and **30** trial records; citation validation, Redis-backed quotas and exportable briefs
- Paired evaluation against a same-model BM25 baseline on **40 held-out** tasks: agent 17/40 vs baseline 19/40, short of the 34/40 target; published case-level results and failure analysis

PharmaEvidence: AI claim review for commercial content | *Python, BM25, DeBERTa NLI, ONNX* — Live | Code
- Checks draft claims against **2,309** passages from versioned KEYTRUDA and WINREVAIR labels with BM25 retrieval, numeric and qualifier rules and a pretrained **INT8 DeBERTa NLI** model running on-device
- Froze **120** authored claims (72 development / 48 held out): NLI **47/48**, rules + NLI **48/48**; the rules caught a tenfold dose error the model accepted

Also built: Field Force Planner *(territory optimisation)* · Footnote · Agent Desk *(AI agents)* · GCC Market Report — GitHub

Experience

Backend Developer Intern, Amaani (Travel-Tech Startup) — Dubai, UAE
BITS Practice School (PS-1) — *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

Research & Publications

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices — 2025 – 2026
First Author; submitted to IEEE MSN 2026 (under review)
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

Technical Skills

AI Solutions & Agents: AI agents, tool calling, RAG, retrieval (BM25, embeddings), natural language inference, structured output, Claude and Gemini APIs
Machine Learning: scikit-learn, TensorFlow/Keras, TensorFlow Lite, INT8 quantization, Transformers.js, ONNX Runtime, held-out evaluation, bootstrap intervals, backtesting
Programming: Python (Pandas, NumPy), SQL (DuckDB, MySQL), TypeScript, Java, C, FastAPI, Flask, REST APIs, Next.js/React
Data & Reporting: data harmonization, Excel, Power BI (DAX), automated reporting, Git, GitHub Actions, pytest, Playwright

Certifications

The Ultimate Job Ready Data Science Course – CodeWithHarry (2026)